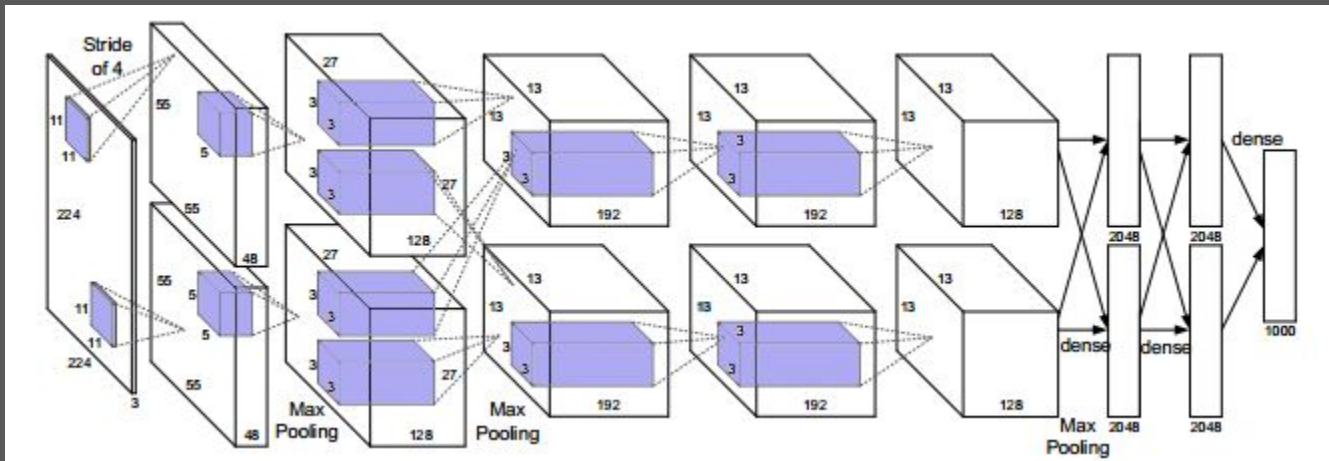
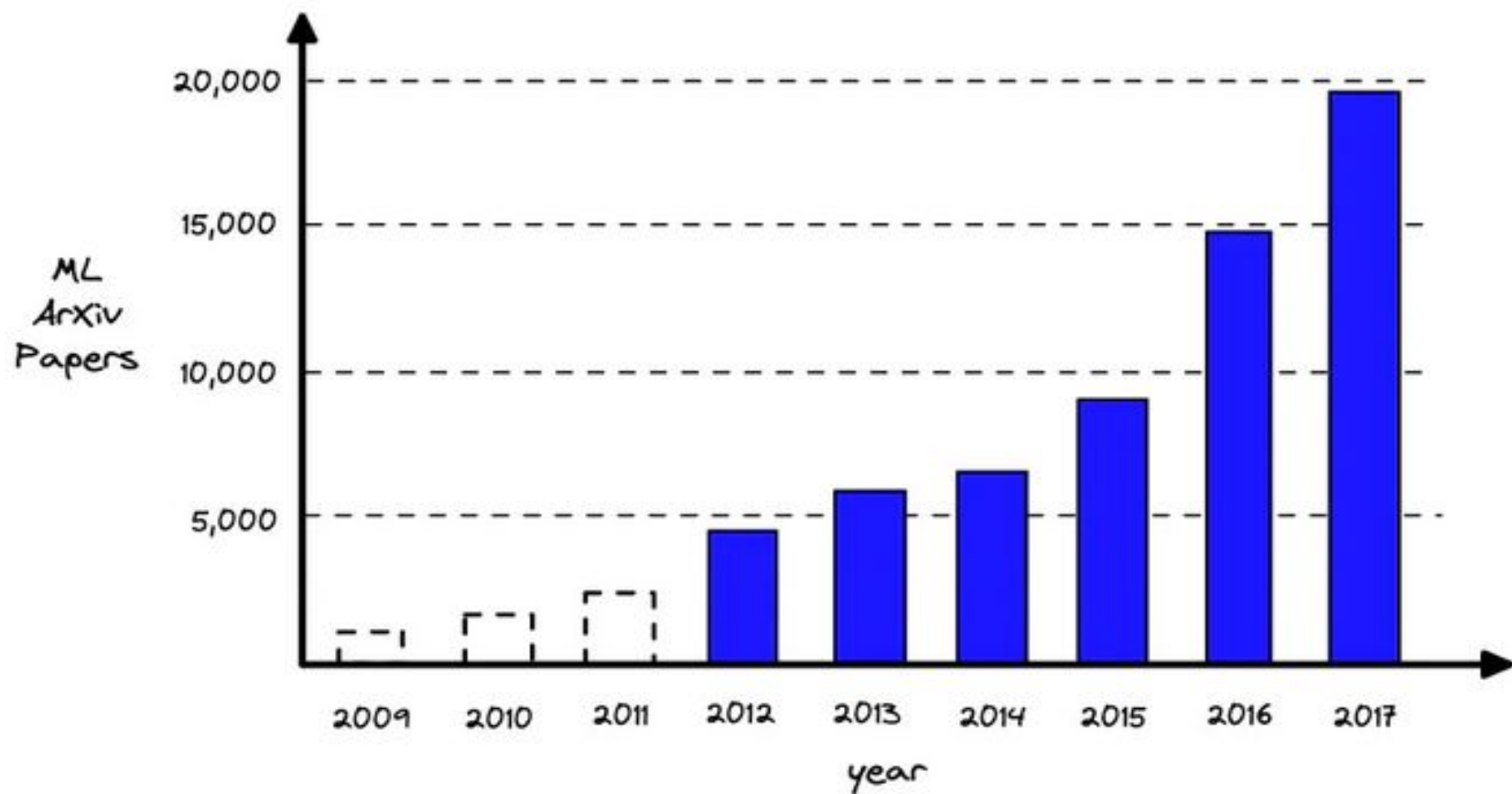
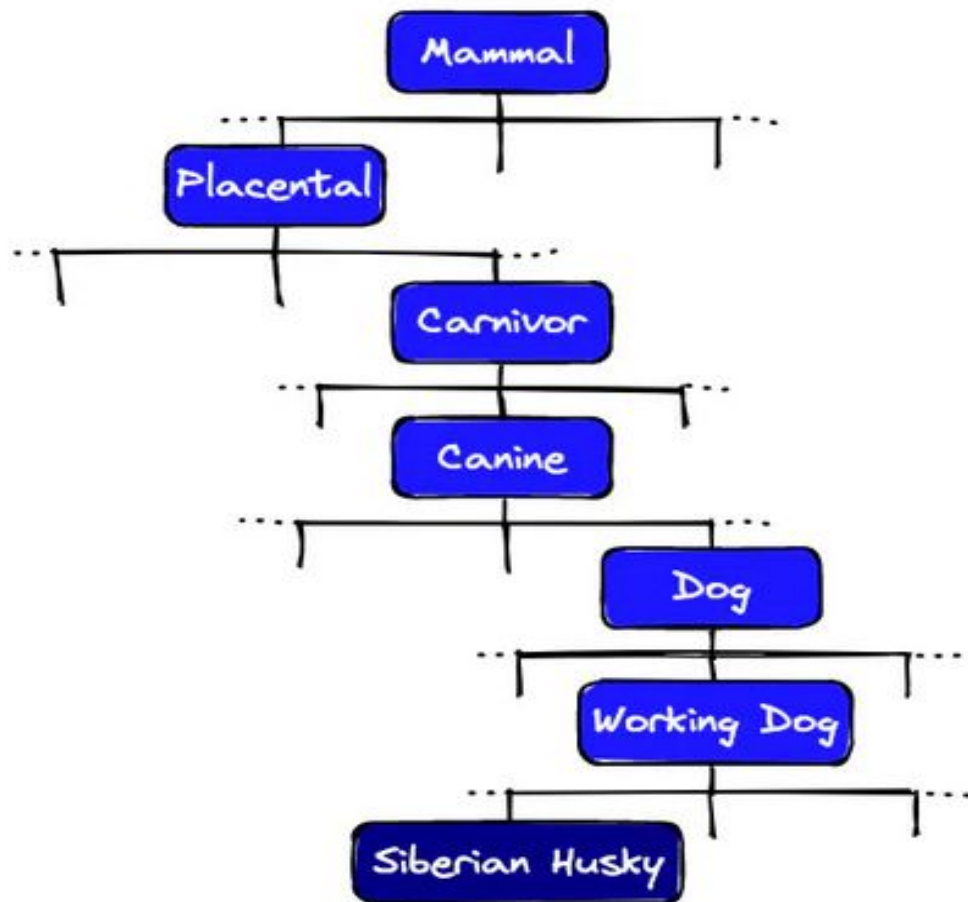


AlexNet: Deep learning made practical





Number of "ML" papers in ArXiv per year [3].



WordNet

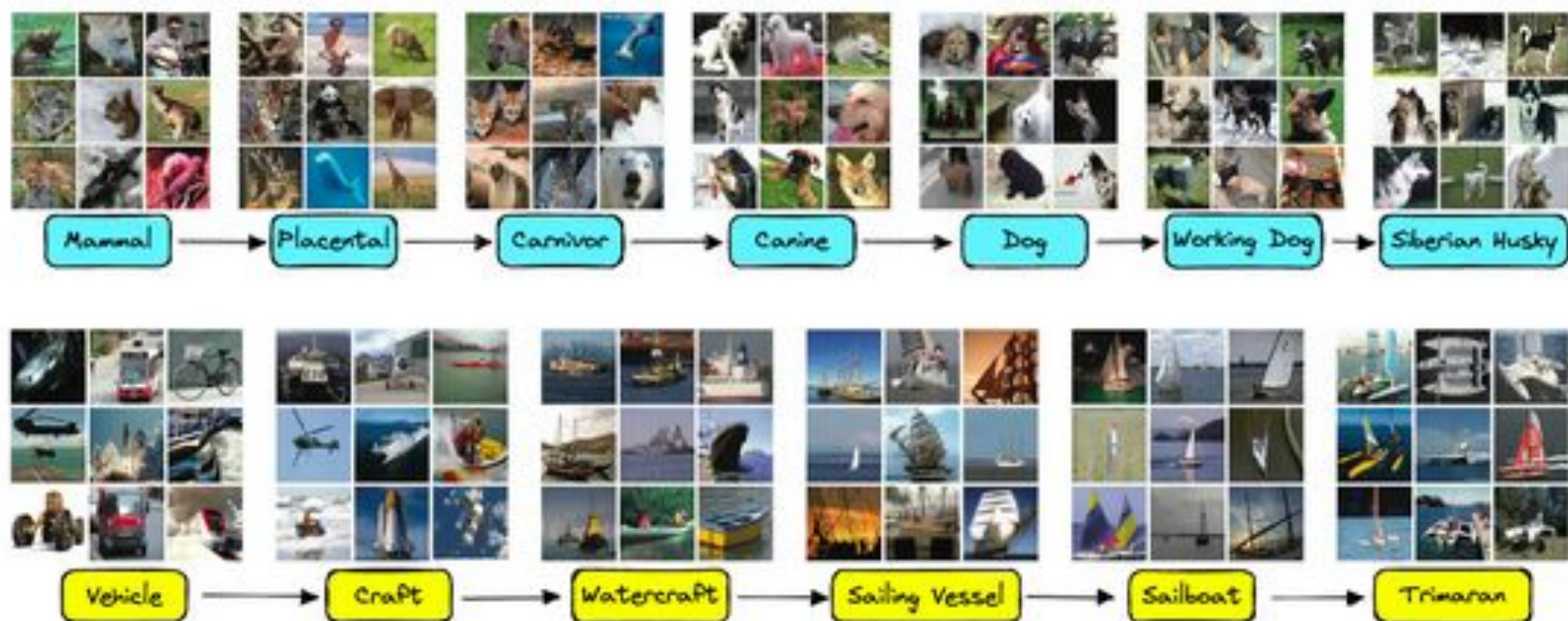
CrowdSourcing

Amazon Mechanical Turk

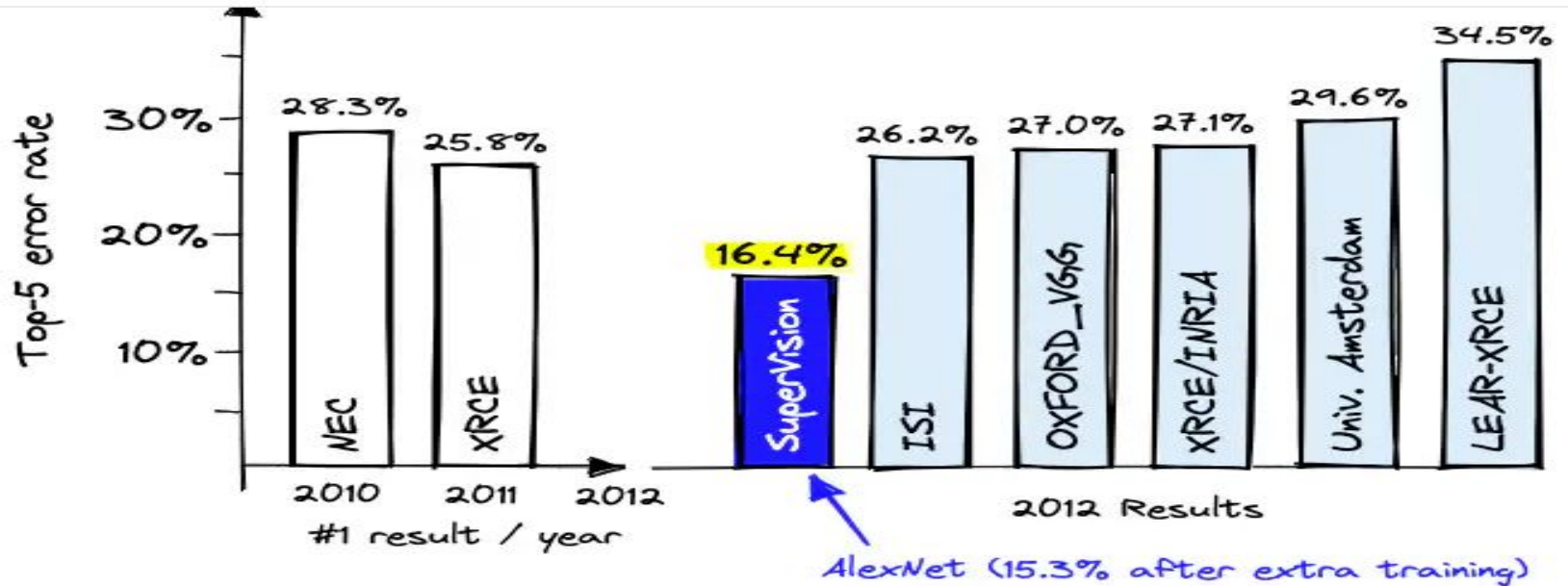
Access a global, on-demand, 24x7 workforce

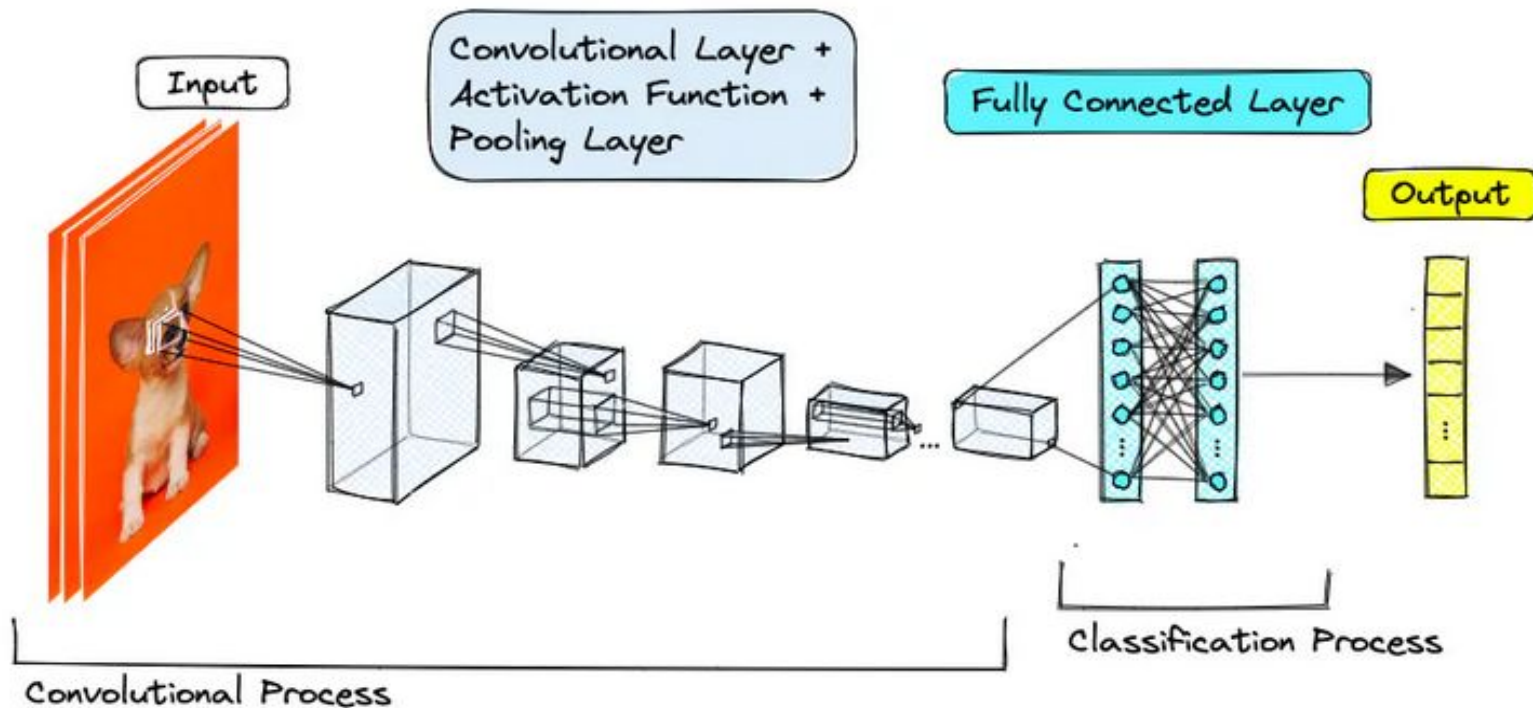


ImageNet

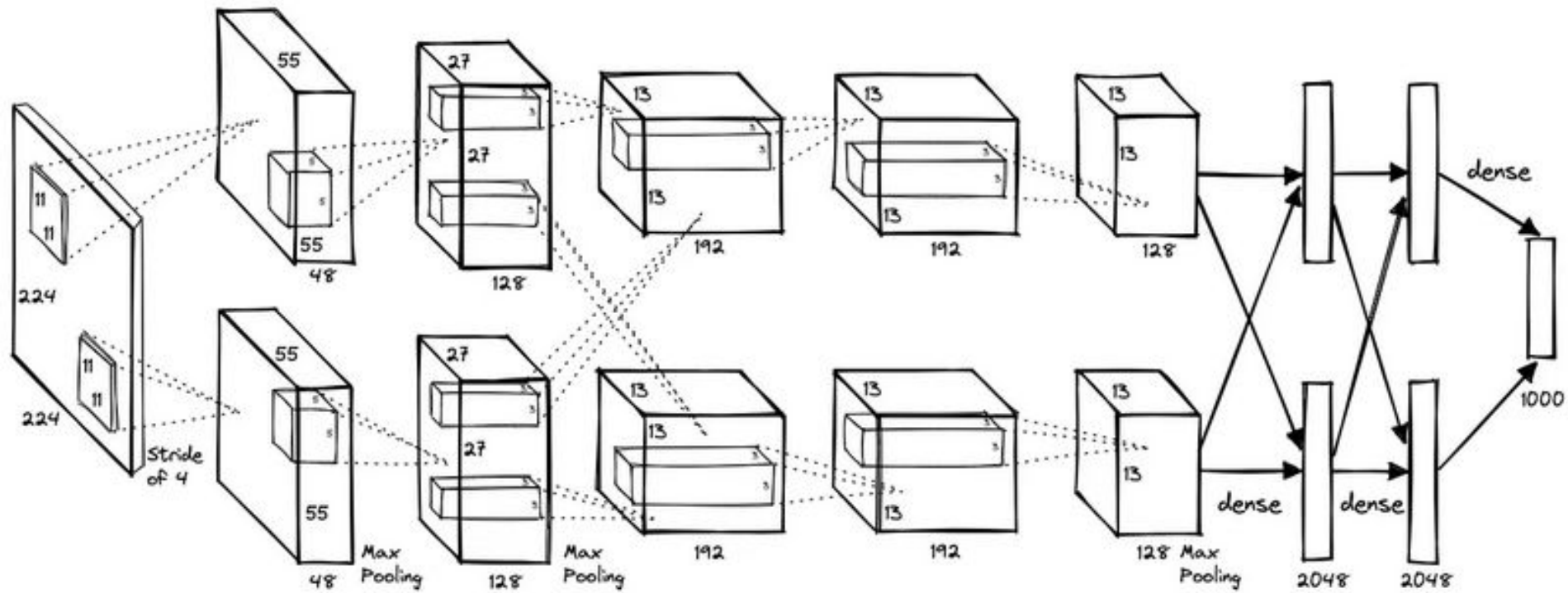


Error percentage dropped substantially

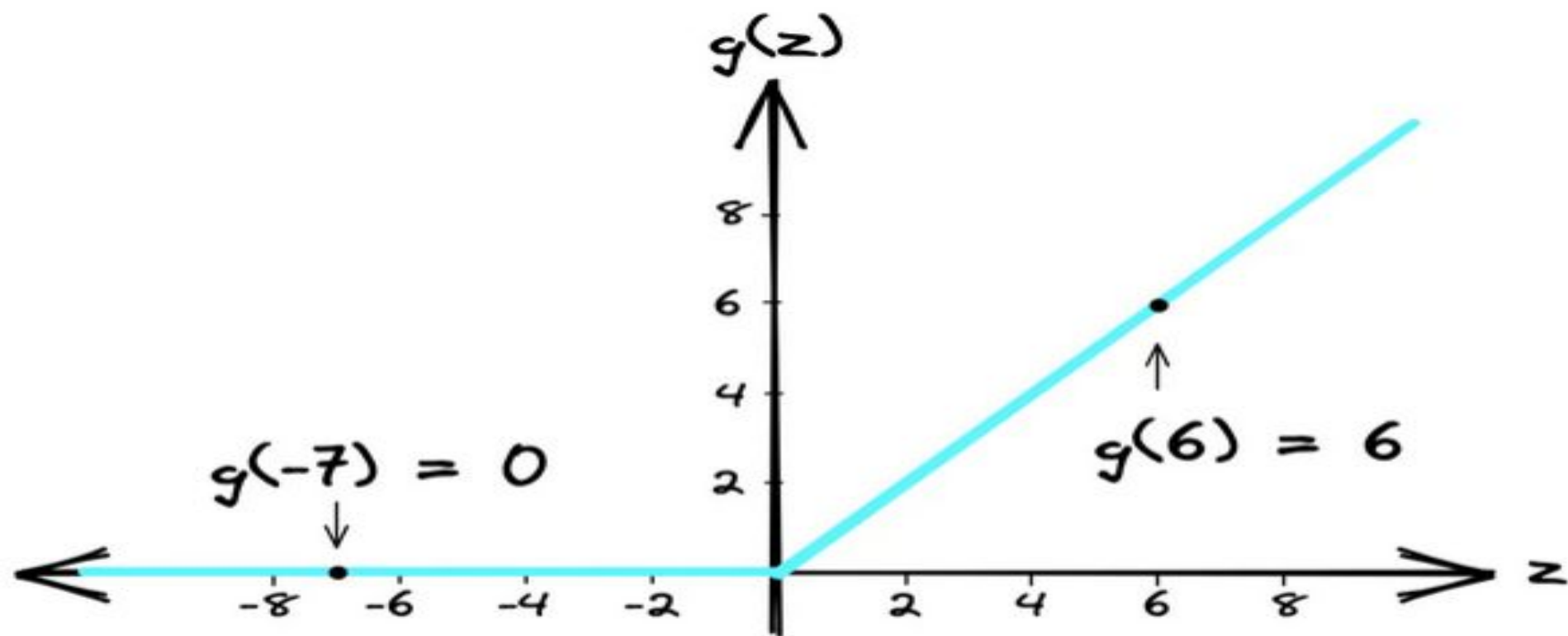




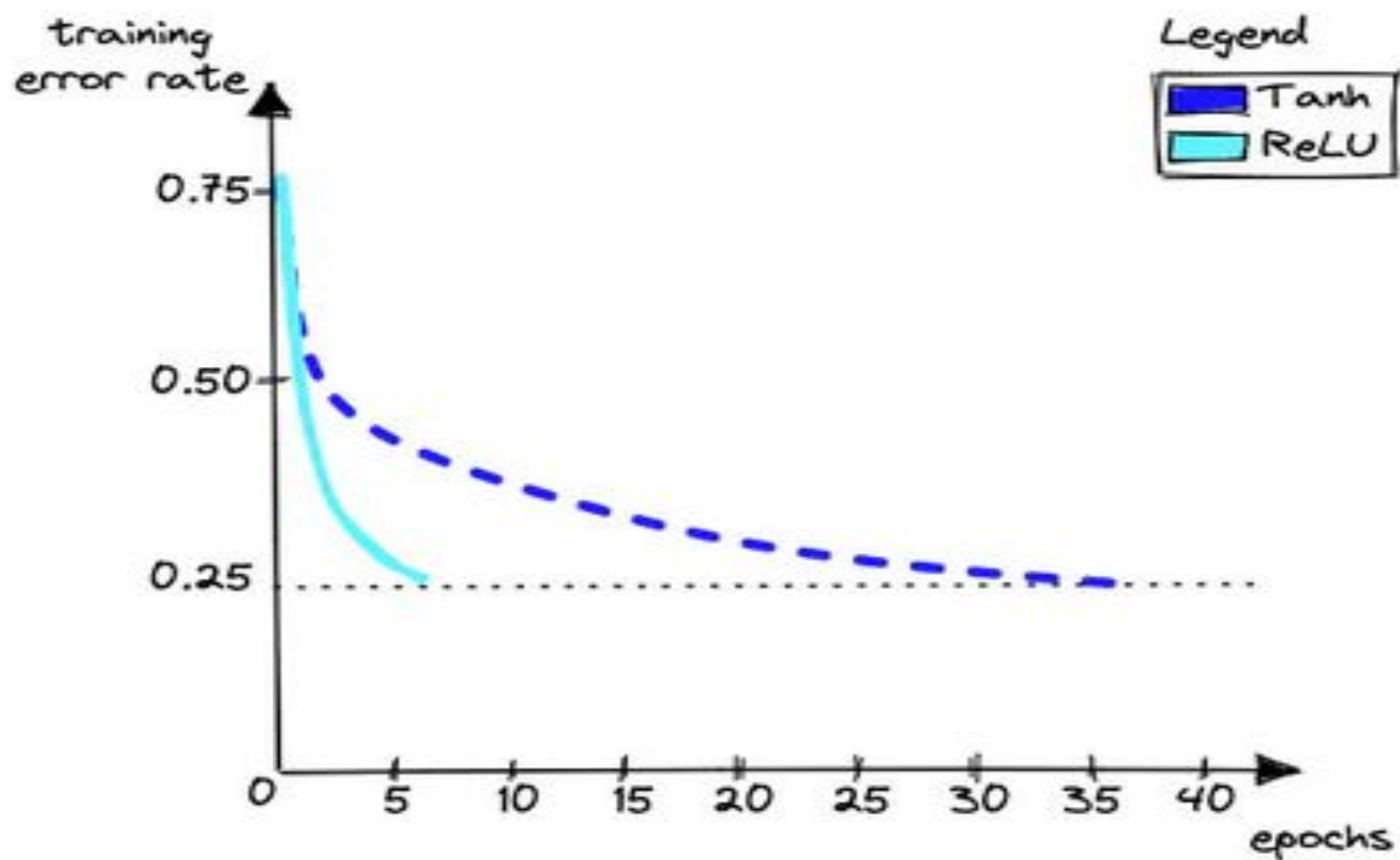
Typical architecture of a CNN.



Network architecture of AlexNet [1].

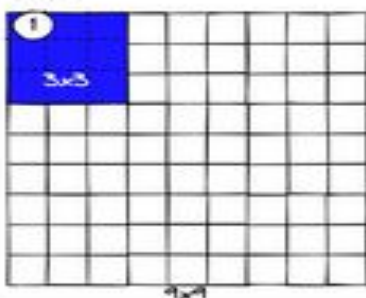


$$g(z) = \max(0; z)$$

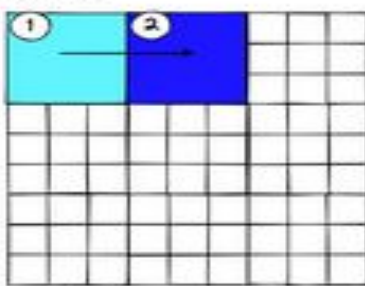


Non-Overlapping Pooling

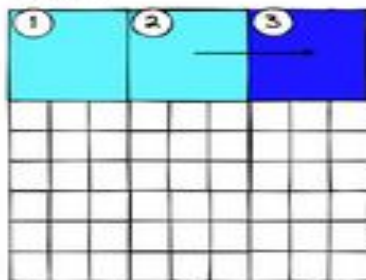
Step 1



Step 2



Step 3

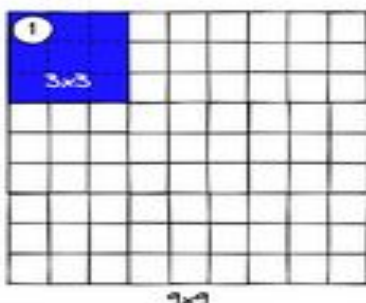


... $\xrightarrow{\text{Output}}$

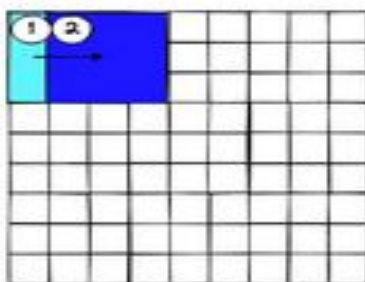


Overlapping Pooling

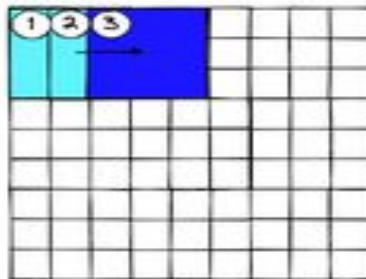
Step 1



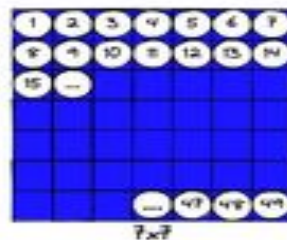
Step 2



Step 3

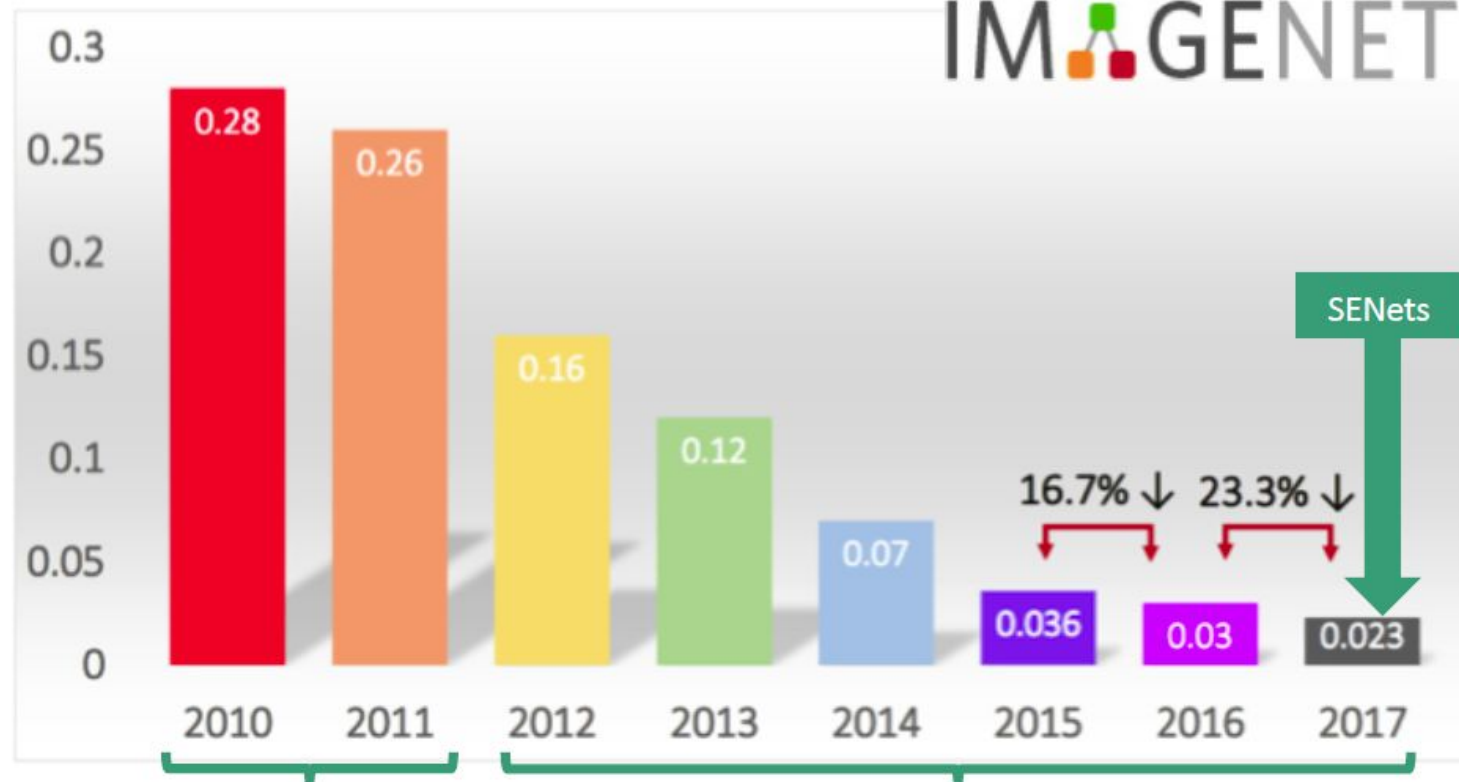


... $\xrightarrow{\text{Output}}$



IMAGENET

Classification Error



Feature Engineering

Convolutional Neural Networks

[Statistics provided by ILSVRC]

AlexNet



ImageNet Large Scale Visual Recognition Challenges

